

King Fahd University of Petroleum and Minerals

College of Computing and Mathematics

Computer Engineering Department

COE 531: Advanced Quantum Computing and Communication

Term 261



Course Information

- Lectures: MW, 8:10-9:25 PM
- Office hours: MW 9:30-10:30 PM for calls on Teams (or send me on Teams and I will try to accommodate your Qs online)

Course Description

Combinatorial optimization. Quantum Optimization. Variational quantum algorithms. Quantum noise models. Quantum error-correction. Quantum fault-tolerance. Quantum communication protocols. Quantum key exchange: BB84. Quantum Internet.

Prerequisites Graduate Level

Course Objectives

The objective of this course is to

- Expose the students to advanced quantum algorithms in NISQ era.
- Introduce the students to theoretical and practical knowledge for building quantum communication protocols.

Learning Outcomes

After taking this course, students will have the ability to

1. Explain the underlying principles of quantum combinatorial optimization problems, and discuss the potential application
2. Understand the correct type of quantum algorithm for each problem
3. Design and analyze variational quantum algorithms
4. Identify the challenges of transferring quantum information
5. Implement and utilize quantum error-correction codes to enhance the reliability of quantum computing system

List of Topics

The following schedule is tentative and subject to changes. More details will be announced in the class and course website/Blackboard.

1. Combinatorial optimization	6
2. Quantum optimization algorithms	6
3. Variational quantum algorithms	6
4. Quantum information theory	3
5. Quantum error-correction and fault-tolerance	9
6. Quantum communication protocols	6
7. Quantum key exchange	6
8. Quantum Internet	3

Textbook

1. Quantum Computing and Information: A Scaffolding Approach

References

1. Combinatorial Optimization

Evaluation

Short Quizzes	15%	
2 Programming Assignments	20%	
Lab Experiment	10%	
Final Exam	30%	(Date TBD by registrar)
Term Project	25%	

Suggested Projects:

- 1- Optimization problems
 - a. Radio Antenna Placement for Network Coverage Optimization
 - b. Autonomous Mission Routing for High-Risk Environments
 - c. Optimized Spectrum and Frequency Assignment
 - d. Optimized Satellite Tasking and Mission Scheduling
 - e. Network Resilience Analysis and Critical Node Identification
- 2- PDEs
 - a. Heat equation
 - b. Poisson equation
 - c. Navier-stoke
 - d. Fisher-KPP
 - e. SIR model

